

Nonracial Transport on Countable Matrix Bundles

An infinite-dimensional subcase of Wirth's question

Autonomous proof candidate for human review

11 August 2026

Status. Candidate partial result, likely valid. The construction covers countable direct products of finite-dimensional detailed-balance systems, hence genuine infinite-dimensional von Neumann algebras with nonracial faithful normal states. It does not cover semigroups mixing the central summands, factors of type II or III, or reference weights.

1 Source question

Wirth [1] constructs a transport metric for tracially symmetric quantum Markov semigroups. At the end of Example 4.19 (source PDF page 26), after recalling the finite-dimensional construction of Carlen and Maas [2], he asks:

How can the construction of the metric $\mathcal{W}$ be generalized to infinite-dimensional quantum Markov semigroups satisfying detailed balance for a nonracial state or weight?

The result below answers the question for decomposable semigroups on a countable matrix bundle. It uses the Carlen–Maas theorem on each fibre, but the infinite product metric, its exact distance formula, and the global entropy-gradient-flow identification require a separate argument.

2 Setup and statement

For every $n \geq 1$, let $\mathcal{M}_n = M_{d_n}(\mathbb{C})$ with normalized trace tr_n , let $\sigma_n \in \mathcal{M}_n$ be a strictly positive density matrix, and let $(P_t^{(n)})_{t \geq 0}$ be a quantum Markov semigroup satisfying detailed balance with respect to σ_n . Write $g^{(n)}$ and $\mathcal{W}_n$ for its Carlen–Maas Riemannian metric and induced transport distance on the strictly positive density matrices $\mathcal{D}_n^+$. Thus the predual evolution $P_{t,*}^{(n)}$ is the $g^{(n)}$ -gradient flow of

$$E_n(r) = \text{tr}_n(r(\log r - \log \sigma_n)).$$

Let

$$\mathcal{M} = \prod_{n \geq 1} \mathcal{M}_n, \quad \phi(a) = \sum_{n \geq 1} w_n \text{tr}_n(\sigma_n a_n),$$

where $w_n > 0$ and $\sum_n w_n = 1$. This is a faithful normal state. It is nonracial as soon as some σ_n is nonscalar. Define the decomposable semigroup $P_t(a) = (P_t^{(n)} a_n)_n$. It is a quantum Markov semigroup on $\mathcal{M}$, is strongly continuous on the predual, and satisfies detailed balance with respect to ϕ , by summing the corresponding fibre identities.

Example 4.19. Let $\mathcal{M}$ be a finite-dimensional von Neumann algebra, tr the normalized trace on $\mathcal{M}$, and (P_t) a quantum Markov semigroup on $\mathcal{M}$. Under the assumption that (P_t) satisfies the quantum detailed balance condition, Carlen and Maas [CM17] defined a Riemannian metric on the space $\mathcal{D}_+(\mathcal{M}, \text{tr})$ of strictly positive density matrices. Let us shortly summarize their construction.

Given $\omega = (\omega_1, \dots, \omega_n) \in \mathbb{R}^n$ and $c = (c_1, \dots, c_n) \in \mathbb{R}^n$ (which are canonically associated with (P_t)) and a density matrix ρ , they define

$$[\rho]_{\omega_j} : L^2(\mathcal{M}, \text{tr}) \longrightarrow L^2(\mathcal{M}, \text{tr}), [\rho]_{\omega_j} = \int_0^1 (e^{-\omega_j/2} L_\rho)^s (e^{\omega_j/2} R_\rho)^{1-s} ds,$$

where L_ρ and R_ρ are the left and right multiplication with ρ on $\mathcal{M}$. Further, $[\rho]_\omega = [\rho]_{\omega_1} \oplus \dots \oplus [\rho]_{\omega_n}$.

The norm of a tangent vector $\dot{\rho}_0$ is defined by

$$g(\dot{\rho}_0, \dot{\rho}_0) = \inf_V \sum_{j=1}^n c_j \langle V_j, [\rho_0]_{\omega_j} V_j \rangle_{L^2(\mathcal{M}, \text{tr})},$$

where the infimum is taken over all V satisfying a continuity equation of the form

$$\dot{\rho}_0 = \text{div}([\rho_0]_\omega V).$$

In the case when (P_t) is tracially symmetric and $\theta = \text{LM}$, one has $c_j = 1$, $\omega_j = 0$ for all $j \in \{1, \dots, n\}$, and it is easily checked that L , R , div etc. coincide with the operations obtained from the first-order differential calculus. Therefore, $[\rho]_0 = \hat{\rho}$ and the distance function induced by g coincides with $\mathcal{W}$.

However, it should be stressed that the class of quantum Markov semigroups satisfying the detailed balance condition is larger than the class of tracially symmetric ones, so we do not fully recover the construction from [CM17]. It is an interesting open question how one can generalize the construction of the metric $\mathcal{W}$ to the case of infinite-dimensional quantum Markov semigroups satisfying the detailed balance condition for a non-tracial state or weight.

Next we collect some basic properties of $\mathcal{W}$. The first one is a sufficient condition to make $\mathcal{W}$ non-degenerate.

Proposition 4.20. *If $\rho_0, \rho_1 \in \mathcal{D}(\mathcal{M}, \tau)$ and $a \in \mathcal{A}_{\text{NM}}$, then*

Figure 1: The finite-dimensional construction and open question in the original paper, source PDF page 26.

A normal state on $\mathcal{M}$ is a sequence $\rho = (p_n r_n)_n$, where $p_n \geq 0$, $\sum_n p_n = 1$, and r_n is a density matrix whenever $p_n > 0$. The numbers p_n are its central masses. Since P_t preserves each central projection, the vector $p = (p_n)$ is invariant. Fix $p_n > 0$ and put

$$\mathcal{D}_p^+ = \{(p_n r_n)_n : r_n \in \mathcal{D}_n^+\}.$$

For $\rho = (p_n r_n)$ and a tangent vector $h = (p_n h_n)$ with $h_n = h_n^*$ and $\text{tr}_n h_n = 0$, define the possibly infinite quadratic form

$$g_\rho^p(h, h) = \sum_{n \geq 1} p_n g_{r_n}^{(n)}(h_n, h_n). \quad (1)$$

Also set

$$\mathcal{W}_p(\rho, \eta)^2 = \sum_{n \geq 1} p_n \mathcal{W}_n(r_n, s_n)^2, \quad \eta = (p_n s_n). \quad (2)$$

Theorem 1 (Countable matrix-bundle extension). *The function $\mathcal{W}_p$ is an extended nondegenerate metric on $\mathcal{D}_p^+$, and it is the length distance associated with the product Riemannian form (1). On the finite-distance, finite-entropy domain, the predual evolution*

$$\rho_t = P_{t,*} \rho_0 = (p_n P_{t,*}^{(n)} r_{n,0})_{n \geq 1}$$

is the gradient flow of the relative entropy

$$\text{Ent}_\phi(\rho) = \sum_{n \geq 1} p_n E_n(r_n) + \sum_{n \geq 1} p_n \log \frac{p_n}{w_n}.$$

More precisely, the Riemannian gradient identity holds for every finite-block tangent vector; whenever the displayed metric-slope and entropy-dissipation sums are finite, it extends by monotone convergence to the full product and the usual energy-dissipation identity is the sum of its fibre identities.

3 Proof intuition

The center records which matrix block a state occupies. A decomposable semigroup cannot transport mass between blocks, so each fixed central-mass fibre is invariant. On that fibre the only possible dynamics are the known finite-dimensional detailed-balance dynamics. The correct kinetic energy is therefore the weighted sum of the fibre kinetic energies, which produces the weighted ℓ_2 product of their distances.

Relative entropy splits in exactly the same way. Its additional classical term measures the discrepancy between the central masses p_n and the reference weights w_n , but this term is constant along the invariant fibre. Consequently its metric gradient is simply the tuple of the fibre gradients.

4 Proof

Lemma 1 (Weighted product length). *Let (X_n, d_n) be length spaces and let $p_n > 0$ with $\sum_n p_n = 1$. On $X = \prod_n X_n$, the formula*

$$d(x, y) = \left(\sum_n p_n d_n(x_n, y_n)^2 \right)^{1/2}$$

defines an extended metric. Its restriction to every finite-distance component is the length metric of the product speed

$$|\dot{\gamma}_t|^2 = \sum_n p_n |\dot{\gamma}_{n,t}|^2.$$

Proof. Separation and symmetry are immediate. Minkowski's inequality in weighted ℓ_2 gives the triangle inequality. For every rectifiable product curve, $d_n(x_n, y_n) \leq \int_0^1 |\dot{\gamma}_{n,t}| dt$; Minkowski's integral inequality then gives

$$d(x, y) \leq \int_0^1 \left(\sum_n p_n |\dot{\gamma}_{n,t}|^2 \right)^{1/2} dt.$$

Conversely choose in each factor a path of length at most $d_n(x_n, y_n) + \epsilon_n$, reparametrize it with constant speed, and choose $\epsilon_n > 0$ so that $\sum_n p_n \epsilon_n^2$ and $\sum_n p_n d_n(x_n, y_n) \epsilon_n$ are arbitrarily small. Their product path has length at most $(\sum_n p_n (d_n(x_n, y_n) + \epsilon_n)^2)^{1/2}$. This proves the reverse inequality by letting the errors tend to zero. $\square$

Applying Lemma 1 to $(\mathcal{D}_n^+, \mathcal{W}_n)$ proves all metric and length assertions in Theorem 1; the infinitesimal product speed is precisely (1).

It remains to identify the gradient flow. Since scalar central weights commute with all matrices,

$$\begin{aligned}\text{Ent}_\phi(\rho) &= \sum_n \text{tr}_n(p_n r_n (\log(p_n r_n) - \log(w_n \sigma_n))) \\ &= \sum_n p_n E_n(r_n) + \sum_n p_n \log \frac{p_n}{w_n}.\end{aligned}$$

On $\mathcal{D}_p^+$ the second sum is constant. If $h = (p_n h_n)$ has only finitely many nonzero coordinates, differentiation gives

$$D \text{Ent}_\phi(\rho)[h] = \sum_n p_n D E_n(r_n)[h_n].$$

By the Carlen–Maas gradient-flow theorem in each block,

$$D E_n(r_n)[h_n] = g_{r_n}^{(n)}(\text{grad}_{g^{(n)}} E_n, h_n), \quad \dot{r}_{n,t} = -\text{grad}_{g^{(n)}} E_n(r_{n,t}).$$

Multiplying by p_n and summing yields

$$D \text{Ent}_\phi(\rho)[h] = g_\rho^p((\text{grad}_{g^{(n)}} E_n)_n, h)$$

and

$$\dot{\rho}_t = -\text{grad}_{g^p} \text{Ent}_\phi(\rho_t) = (p_n \dot{r}_{n,t})_n.$$

This proves the classical Riemannian identity on the finite-block core.

For a general product trajectory for which the sums are finite, the fibre energy-dissipation identities are nonnegative identities. Summing their first N terms and sending $N \rightarrow \infty$ gives, by monotone convergence, the global identity and hence the asserted gradient-flow statement on its natural finite-energy domain. Strong continuity of the predual product semigroup follows independently from dominated convergence, since each predual map is an L^1 contraction. This completes the proof.

5 Scope, full-upgrade obstruction, and verification

The construction is not merely a finite direct sum: with infinitely many blocks, $\mathcal{M}$ is infinite dimensional and generally nonseparable in operator norm, while ϕ can be genuinely nontracial. Equation (2) may take the value $+\infty$, exactly as Wirth’s transport metric may.

Two upgrades were tested. The same product argument extends formally to measurable fields $L^\infty(X; M_d)$, but an exact length-space statement needs a measurable choice of almost-minimizing fibre paths; this was not inserted as an unproved selection lemma. More importantly, the derivation theorem for arbitrary KMS-symmetric semigroups [3] does not by itself supply the operator-valued logarithmic-mean chain rule needed to identify the Onsager operator on Haagerup L^1 densities. In type III algebras such densities are not bounded elements of $\mathcal{M}$, so the finite-matrix left/right functional calculus used above cannot simply be repeated. Thus no claim is made for the full question.

The proof is purely analytic. The verifier should focus on (i) the intended notion of detailed balance in the source, (ii) the domain on which an infinite sum of fibre energy-dissipation identities is finite, and (iii) whether a standard reference already records this countable-product closure explicitly.

References

- [1] M. Wirth, *A Noncommutative Transport Metric and Symmetric Quantum Markov Semigroups as Gradient Flows of the Entropy*, arXiv:1808.05419 (2018).
- [2] E. A. Carlen and J. Maas, *Gradient flow and entropy inequalities for quantum Markov semigroups with detailed balance*, J. Funct. Anal. 273 (2017), 1810–1869; arXiv:1609.01254.
- [3] M. Vernooij and M. Wirth, *Derivations and KMS-Symmetric Quantum Markov Semigroups*, arXiv:2303.15949 (2023).